Machine learning foundations

Unit 1 · reference sheet · AI/ML in 35 topics

The four objects

$X \in \mathbb{R}^{n \times d}$, $y \in \mathbb{R}^n$, f_θ, L
$\theta^* = \arg\min_\theta \frac{1}{n} \sum_i L(f_\theta(x_i), y_i)$

Diagnosis table

Train / Val	Diagnosis	Lever
high / high	underfit	capacity, features, epochs
low / high	overfit	data, reg., capacity, early stop
high / high (flat)	wrong class	change model family

Linear & logistic

$\hat{y} = w \cdot x + b$ closed form $(X^\top X)^{-1} X^\top y$
$\sigma(z) = \dfrac{1}{1+e^{-z}}$ $z = \log \dfrac{p}{1-p}$
Boundary: $w \cdot x + b = 0$ (always a hyperplane)
BCE $= -[y \log p + (1-y) \log(1-p)]$
Gradient: $\partial L / \partial w = (p - y)x$

Trees & ensembles

Gini $= 1 - \sum_k p_k^2$ $H = -\sum_k p_k \log p_k$
Bagging = variance; boosting = bias.
Trees cannot extrapolate: a leaf predicts a training average.

Gotchas

- Accuracy on 1% positives: 99% by predicting "no".
- Scaling before the split leaks test statistics.
- Impurity importance favours high-cardinality features.

This unit at a glance

#	Topic	You can
4	Supervised learning as function fitting	write down (X, y, f_θ, L) for any task and say what is being learned
5	Generalization	diagnose over- vs underfitting from train/test curves and pick the fix
6	Linear & logistic regression	derive a logistic decision boundary; explain why squared error is wrong for cl
7	Trees & ensembles	explain axis-aligned splits, bagging vs boosting, and when trees beat linear m

If you see this → do that

Signal	Move
Train high, val low	regularise
Both high	more capacity
Classes imbalanced	PR / AUC, not accuracy
Boundary is curved	hidden layer or tree
Need to defend a decision	linear or small tree

Numbers worth memorising

BCE at $p = 0.5$: 0.693 (the "stuck" loss)
AUC baseline for 1% positives: 0.5, accuracy baseline 99%
80/20 gap > 10 points ⇒ overfit before anything else
$\sigma(2) = 0.88$, $\sigma(4) = 0.98$, $\sigma(-2) = 0.12$